

ENGINEERING REBOOT 2026

Week 03 – Day 13 Field Manual

Dictionaries & Structured Data

Version 1.1 – Re-Baselined Edition – Publication Edition

Status: LOCKED

PHASE 0 – MISSION BRIEF

Objective

Learn how software systems represent real-world entities using structured key-value data.

Lists allow storage of multiple values.

Dictionaries allow storage of related attributes describing a single entity.

Day 13 introduces the second major Python data structure: Dictionary

Core Concepts

- dictionaries
- keys
- values
- key-value pairs
- structured records
- data modeling

Engineering Applications

User Record

```
user = {"name": "Alice", "age": 25, "city": "New York"}
```

Server Configuration

```
server = {"hostname": "web01", "ip": "192.168.1.10", "status": "online"}
```

Home Assistant Entity

```
light = {"entity_id": "light.living_room", "state": "on", "brightness": 80}
```

Learning Objectives

- Create dictionaries
- Access values using keys
- Display structured information
- Model real-world entities
- Apply dictionaries to practical software problems

Deliverable Inventory

1. dictionary_basics.py
2. student_record.py
3. inventory_dictionary.py
4. Notes.md
5. Journal.md
6. Evidence Package

PHASE 1 – DICTIONARY BASICS

Filename

dictionary_basics.py

Implement

Create:

```
person = {  
    "name": "Alice",  
    "age": 25,  
    "city": "New York"  
}
```

Display:

- Name
- Age
- City

Validation Matrix

Expected Output

Name: Alice
Age: 25
City: New York

Evidence Checkpoint

dictionary_basics-Code.png
dictionary_basics-Output.png

Completion Checkpoint

- Exercise Completed
- Validation Completed
- Evidence Captured

PHASE 2 – STUDENT RECORD

Filename

student_record.py

Implement

Create:

```
student = {  
    "name": "John",  
    "course": "Python Bootcamp",  
    "score": 95  
}
```

Display:

- Student Name
- Course
- Score

Validation Matrix

Expected Output

Student Name: John
Course: Python Bootcamp
Score: 95

Evidence Checkpoint

student_record-Code.png
student_record-Output.png

Completion Checkpoint

- Exercise Completed
- Validation Completed
- Evidence Captured

PHASE 3 – INVENTORY DICTIONARY

Filename

inventory_dictionary.py

Implement

Create:

```
inventory = {  
    "item": "Laptop",  
    "quantity": 15,  
    "location": "Warehouse A"  
}
```

Display:

- Item
- Quantity
- Location

Validation Matrix

Expected Output

Item: Laptop
Quantity: 15
Location: Warehouse A

Evidence Checkpoint

inventory_dictionary-Code.png
inventory_dictionary-Output.png

Completion Checkpoint

- Exercise Completed
- Validation Completed
- Evidence Captured

PHASE 4 – NOTES

Create

03-Week-03/Day-13/Notes/Notes.md

Required Sections

- Concepts Learned
- Key Commands
- Key Observations
- Lessons Learned
- Questions For Future Study

Suggested Topics

- What is a dictionary?
- How are keys different from indexes?
- When should dictionaries be used?
- How do dictionaries model real-world data?

Template

Concepts Learned

Key Commands

Key Observations

Lessons Learned

Questions For Future Study

Completion Checkpoint

- Notes Completed

PHASE 5 – JOURNAL

Create

03-Week-03/Day-13/Journal/Journal.md

Required Sections

- Summary
- Challenges Encountered
- Solutions Applied
- Confidence Assessment
- Reflection
- Tomorrow Preparation

Suggested Reflection

How does a dictionary represent information differently from a list?

Template

Summary

Challenges Encountered

Solutions Applied

Confidence Assessment

Reflection

Tomorrow Preparation

Completion Checkpoint

■ Journal Completed

PHASE 6 – GIT OPERATIONS

Execute

```
git status
git add .
git commit -m "Complete Day 13 dictionary exercises"
git push
git fetch
git status
git branch -vv
```

Preferred End State

Your branch is up to date with 'origin/main'
nothing to commit, working tree clean

Evidence Checkpoint

Git-Operations.png

Must Show

```
git add .
git commit
git push
git fetch
git status
git branch -vv
```

Completion Checkpoint

- Git Operations Completed
- Repository Synchronized
- Evidence Captured

END-OF-DAY SUCCESS CRITERIA

- dictionary_basics.py completed
- student_record.py completed
- inventory_dictionary.py completed
- All validation completed
- Exercise evidence captured
- Notes completed
- Journal completed
- Git workflow completed
- Git-Operations.png captured
- Remote synchronization verified
- Working tree clean

MISSION COMPLETE ✓

DAY 14 PREVIEW

Topic

Dictionary Traversal & Processing

Expected Deliverables

- employee_directory.py
- inventory_report.py
- student_grades.py

Core Concepts

- dictionary traversal
- keys()
- values()
- items()
- structured processing

Engineering Focus

Processing and reporting structured records using dictionary data.